

**ME 2261 - Embedded Systems Log Book**USART**Initializing serial port**

Set USART for asynchronous mode

Set USART communication parameters (data bits, stop bits, parity bit)

Set baud rate

Enable transmitter and receiver

Initialize serial port of ATmega32A to baud rate 9600 bps, no parity, 1 stop bit, 8 data bits.

Assume a clock speed of 8 MHz.

**Sending a character**

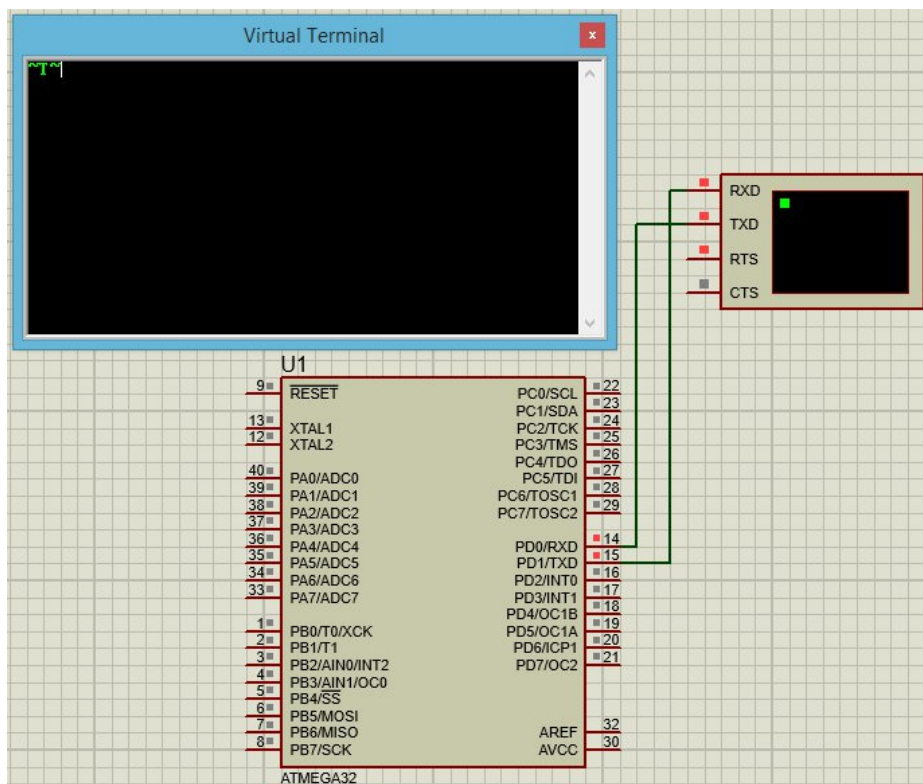
Has UDRE flag (Register UCSRA) been set to 1?

Yes, Write the character to register to UDR for transmission

**Receiving a character**

Has RXC flag (Register UCSRA) set to 1?

Yes, Read the received character from register UDR

Task 1

```

1  #define F_CPU 8000000UL
2  #include <avr/io.h>
3
4  void USART_init(void){
5      // Asynchronous mode, no parity, 1 stop bit, 8 data bits
6      UCSRC = 0b1000110;
7      // Normal speed, disable multi-proc
8      UCSRA = 0b00000000;
9      // Baud rate 9600 bps, assuming 8 MHz clock
10     UBRRL = 0x33;
11     UBRRH = 0x00;
12     // Enable Tx and Rx, disable interrupts
13     UCSRB = 0b00011000;
14 }
15
16 void USART_send(unsigned char data){
17     // Wait until UDRE flag = 1
18     while ((UCSRA & (1<<UDRE))==0x00); // while (!(UCSRA & (1<<UDRE)));
19     // Read the received char from UDR
20     UDR = data;
21 }
22
23 unsigned char USART_receive(void){
24     // Wait until RXC flag = 1
25     while ((UCSRA & (1<<RXC))==0x00); // while (!(UCSRA & (1<<RXC)));
26     // Write UDR to char for receiving
27     return UDR;
28 }
29
30 int main(void)
31 {
32     unsigned char data;
33     USART_init(); // Initializing serial port
34     data = USART_receive();
35     USART_send('~');
36     USART_send(data);
37     USART_send('~');
38 }

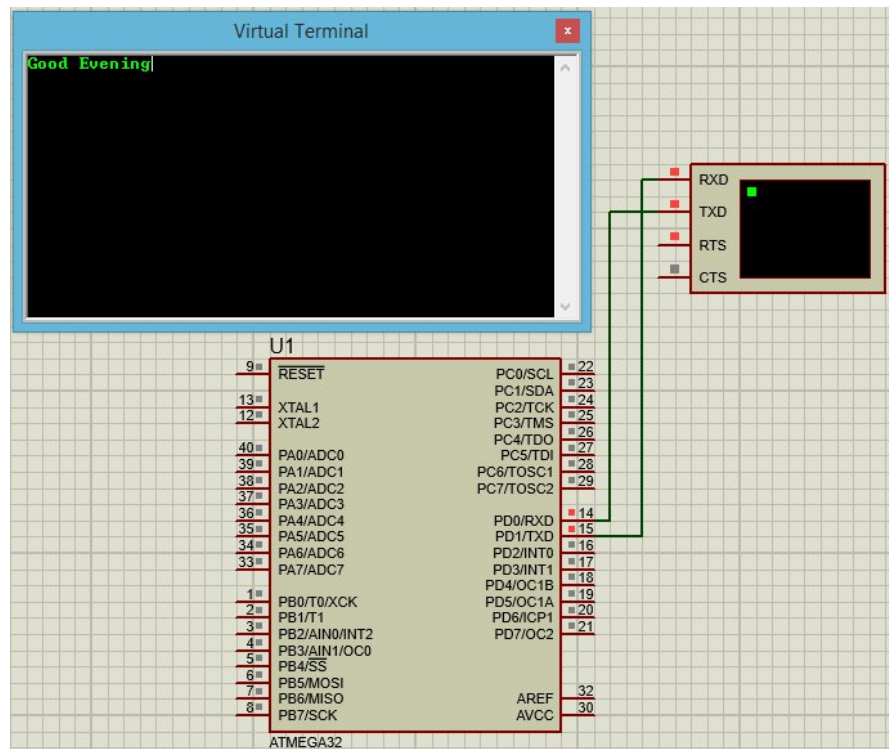
```

## Task 2

Using USART communication interface, write a program to display a message according to the received character.

- If the character 'M' is received, send the message "Good Morning".
- If the character 'A' is received, send the message "Good Afternoon".
- If the character 'E' is received, send the message "Good Evening".

```
int main(void)
{
    unsigned char data;
    USART_init();
    data = USART_receive();
    if (data == 'M'){
        USART_send('G');
        USART_send('o');
        USART_send('o');
        USART_send('d');
        USART_send(' ');
        USART_send('M');
        USART_send('o');
        USART_send('r');
        USART_send('n');
        USART_send('i');
        USART_send('n');
        USART_send('g');
    }
    if (data == 'A'){
        USART_send('G');
        USART_send('o');
        USART_send('o');
        USART_send('d');
        USART_send(' ');
        USART_send('A');
        USART_send('f');
        USART_send('t');
        USART_send('e');
        USART_send('r');
        USART_send('n');
        USART_send('o');
        USART_send('o');
        USART_send('n');
    }
    if (data == 'E'){
        USART_send('G');
        USART_send('o');
        USART_send('o');
        USART_send('d');
        USART_send(' ');
        USART_send('E');
        USART_send('v');
        USART_send('e');
        USART_send('n');
        USART_send('i');
        USART_send('n');
        USART_send('g');
    }
}
```



### Task 3

Send data from one microcontroller to another microcontroller and display the received message on the LCD screen. (Connect TXD of one to RXD of another using wire)

Follow the same message format as Task 2.

```
void USART_send(unsigned char data){
    // Wait until UDRE flag = 1
    while ((UCSRA & (1<<UDRE))==0x00);    // while (!(UCSRA & (1<<UDRE)));
    // Read the received char from UDR
    UDR = data;
}

unsigned char USART_receive(void){
    // Wait until RXC flag = 1
    while ((UCSRA & (1<<RXC))==0x00);    // while (!(UCSRA & (1<<RXC)));
    // Write UDR to char for receiving
    return UDR;
}

int main(void)
{
    unsigned char data;
    USART_init();
    while(1){
        data = USART_receive();
        USART_send(data);
    }
}
```

---

```
unsigned char USART_receive(void){
    // Wait until RXC flag = 1
    while ((UCSRA & (1<<RXC))==0x00);    // while (!(UCSRA & (1<<RXC)));
    // Write UDR to char for receiving
    return UDR;
}

int main(void)
{
    DDRA = 0xFF;
    DDRB = 0xFF;
    Lcd8_Init();
    unsigned char data;
    USART_init();
    while (1){
        data = USART_receive();
        Lcd8_Set_Cursor(1,0);
        if (data == 'M')
        {
```

```

        Lcd8_Clear();
        Lcd8_Write_String("Good Morning");
    }
    if (data == 'A')
    {
        Lcd8_Clear();
        Lcd8_Write_String("Good Afternoon");
    }
    if (data == 'E')
    {
        Lcd8_Clear();
        Lcd8_Write_String("Good Evening");
    }
    _delay_ms(1000);
}
}

```

